EXP1:

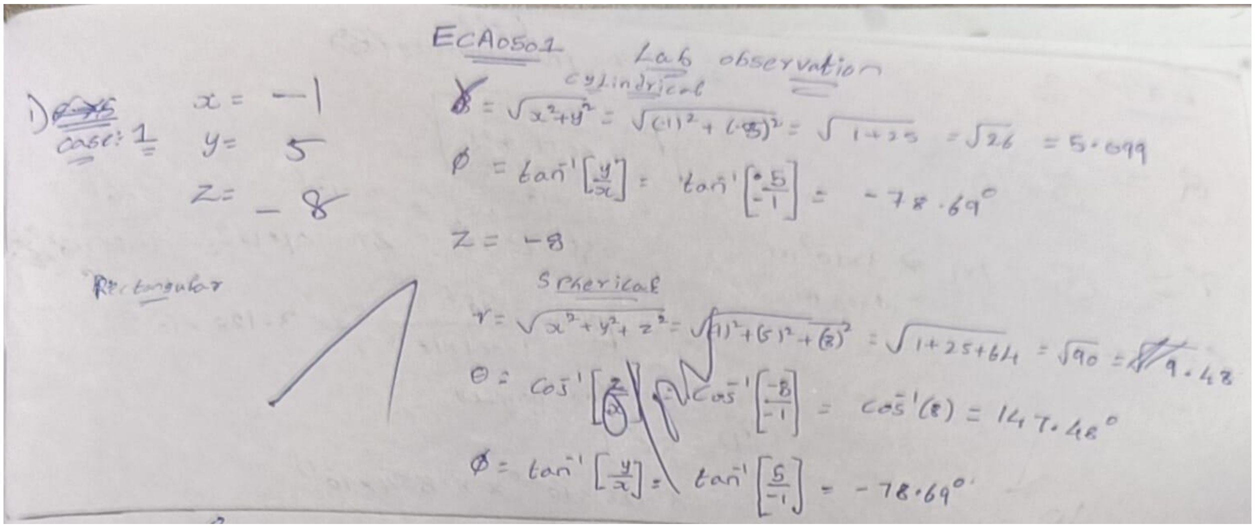
Variable Transformation between Cartesian, Cylindrical and Spherical coordinate system to obtain the location of any object.

TEST CASE 1

Code:

```
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');
r=sqrt(x^2+y^2);
pi=atand(y/x);
disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=',[r pi z]);
r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
pi=atand(y/x);
disp('Cartesian to spherical coordinate system of s(r,theta,pi)=',[r theta pi]);
```

Calculation:



ECA0501 Lab observation

1) Case: 1 $x = -1$, $y = 5$, $z = -8$

Cylindrical

$r = \sqrt{x^2+y^2} = \sqrt{(-1)^2+(5)^2} = \sqrt{1+25} = \sqrt{26} = 5.099$

$\phi = \tan^{-1}\left[\frac{y}{x}\right] = \tan^{-1}\left[\frac{5}{-1}\right] = -78.69^\circ$

$z = -8$

Rectangular

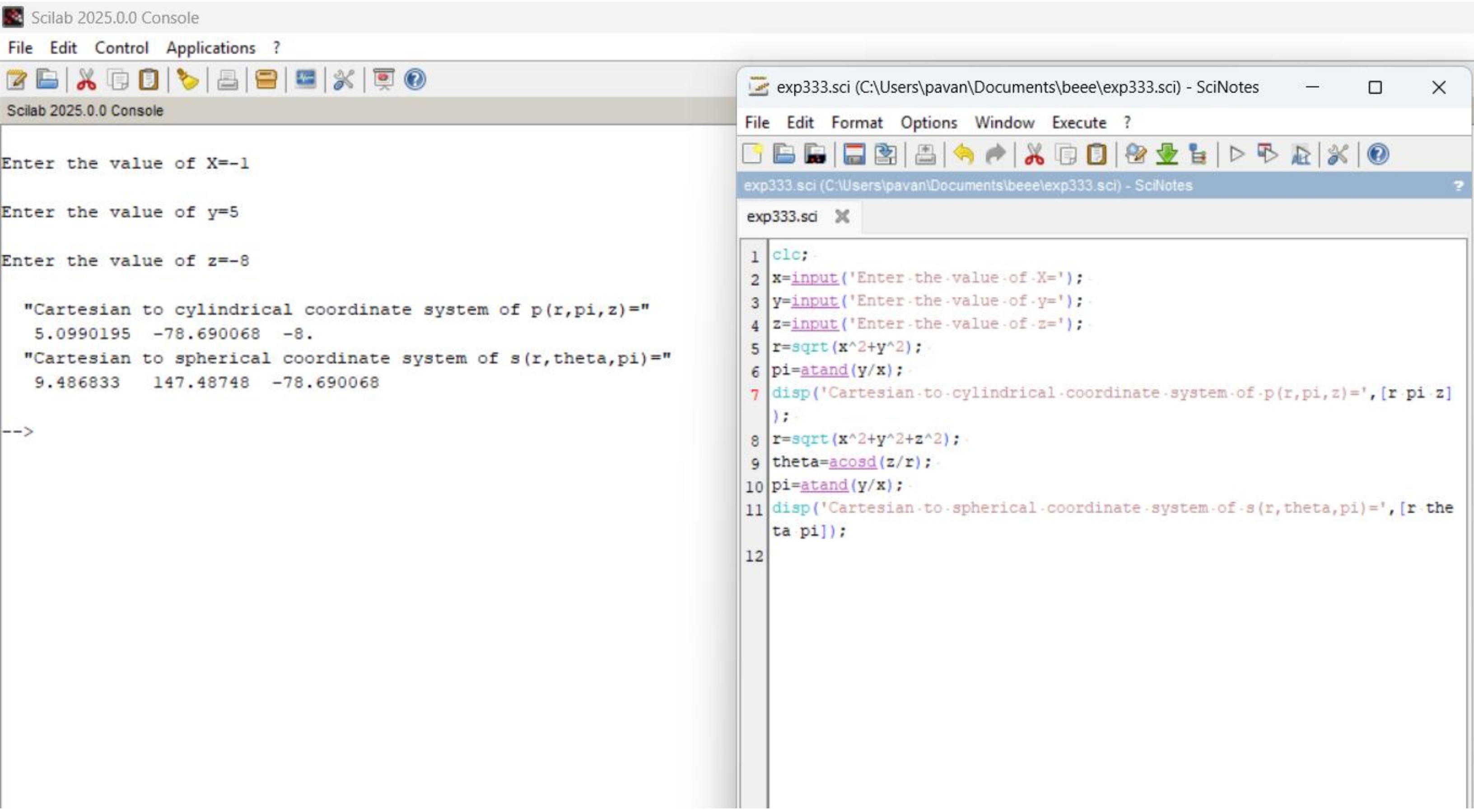
Spherical

$r = \sqrt{x^2+y^2+z^2} = \sqrt{(-1)^2+(5)^2+(8)^2} = \sqrt{1+25+64} = \sqrt{90} = 9.48$

$\theta = \cos^{-1}\left[\frac{z}{r}\right] = \cos^{-1}\left[\frac{-8}{-1}\right] = \cos^{-1}(8) = 147.48^\circ$

$\phi = \tan^{-1}\left[\frac{y}{x}\right] = \tan^{-1}\left[\frac{5}{-1}\right] = -78.69^\circ$

OUTPUT:


Scilab 2025.0.0 Console
File Edit Control Applications ?
Scilab 2025.0.0 Console
Enter the value of X=-1
Enter the value of y=5
Enter the value of z=-8
"Cartesian to cylindrical coordinate system of p(r,pi,z)="
5.0990195 -78.690068 -8.
"Cartesian to spherical coordinate system of s(r,theta,pi)="
9.486833 147.48748 -78.690068
-->
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
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exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');
r=sqrt(x^2+y^2);
pi=atand(y/x);
disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=',[r pi z]);
r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
pi=atand(y/x);
disp('Cartesian to spherical coordinate system of s(r,theta,pi)=',[r theta pi]);

TEST CASE 2

Code:

```
clc;
r=input('Enter the value of r=');
pi=input('Enter the value of pi=');
z=input('Enter the value of z=');
x=r*cosd(pi);
y=r*sind(pi);
z=z;
disp('cylindrical to cartesian coordinate system of p(x,y,z)=',[x y z]);
r=sqrt(r^2+z^2);
pi=pi;
theta=acosd(z/r);
disp('cylindrical to spherical coordinate system of s(r,pi,theta)=',[r pi theta]);
```

Calculation:

2) case:-2

$\rho = 5$

$\phi = 30^\circ$

$z = -5$

cylindrical

Rectangular

$x = \rho\cos\phi = 5\cos(30^\circ) = 4.33.$

$y = \rho\sin\phi = 5\cos(30^\circ) = 2.5$

$z = z = -5$

Spherical

$r = \sqrt{\rho^2+z^2} = \sqrt{(5)^2+(-5)^2} = \sqrt{25+25} = \sqrt{50} = 7.07$

$\phi = \phi = 30^\circ$

$\theta = \tan^{-1}\left[\frac{\rho}{z}\right] = \tan^{-1}\left[\frac{5}{-5}\right] = \tan^{-1}(-1) = -45$

Output:

```
Enter the value of r=5

Enter the value of pi=30

Enter the value of z=-5

  "cylindrical to cartesian coordinate system of p(x,y,z)="
   4.330127   2.5   -5.
  "cylindrical to spherical coordinate system of s(r,pi,theta)="
   7.0710678   30.   135.
```

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

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exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

exp333.sci

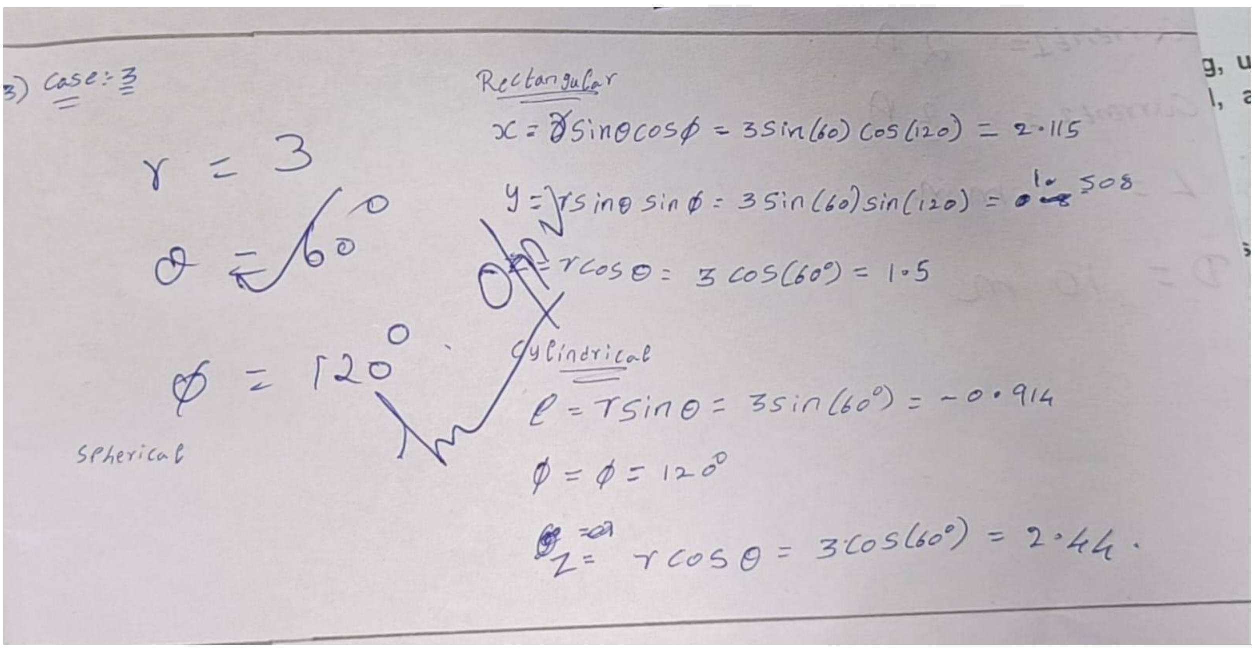
```
clc;
clear;
r=input('Enter the value of r=');
pi=input('Enter the value of pi=');
z=input('Enter the value of z=');
x=r*cosd(pi);
y=r*sind(pi);
z=z;
disp('cylindrical to cartesian coordinate system of p(x,y,z)=',[x y z]);
r=sqrt(r^2+z^2);
pi=pi;
theta=acosd(z/r);
disp('cylindrical to spherical coordinate system of s(r,pi,theta)=',[r pi theta]);
```

TEST CASE 3

Code:

```
clc;

r=input('Enter the value of r=');

pi=input('Enter the value of pi=');

theta=input('Enter the value of theta=');

x=r*sind(theta)*cos(pi);

y=r*sind(theta)*sin(pi);

z=r*cosd(theta);

disp('spherical to cartesian coordinate system of p(x,y,z)=',[x y z]);

p=r*sin(theta);

pi=pi;

z=r*cos(pi);

disp('spherical to cylindrical coordinate system of s(p,pi,z)=',[p pi z]);
```

Calculation:



3) Case:- 3

$r = 3$

$\theta = 60^\circ$

$\phi = 120^\circ$

Spherical

Rectangular

$x = r\sin\theta\cos\phi = 3\sin(60)\cos(120) = 2.115$

$y = r\sin\theta\sin\phi = 3\sin(60)\sin(120) = 1.508$

$z = r\cos\theta = 3\cos(60^\circ) = 1.5$

Cylindrical

$\rho = r\sin\theta = 3\sin(60^\circ) = -0.914$

$\phi = \phi = 120^\circ$

$z = r\cos\theta = 3\cos(60^\circ) = 2.44.$

Output:

```
Enter the value of r=3

Enter the value of pi=60

Enter the value of theta=120

  "spherical to cartesian coordinate system of p(x,y,z)="
  -2.4744415  -0.7919212  -1.5
  "spherical to cylindrical coordinate system of s(p,pi,z)="
   1.7418336   60.  -2.8572389
```

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exp333.sci

```
clc;
r=input('Enter the value of r=');
pi=input('Enter the value of pi=');
theta=input('Enter the value of theta=');
x=r*sind(theta)*cos(pi);
y=r*sind(theta)*sin(pi);
z=r*cosd(theta);
disp('spherical to cartesian coordinate system of p(x,y,z)=',[x y z]);
p=r*sin(theta);
pi=pi;
z=r*cos(pi);
disp('spherical to cylindrical coordinate system of s(p,pi,z)=',[p pi z]);
```